Gradient Descent Derivation

Linear regression model

$$f_{w,b}(x) = wx + b$$

Cost function

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^{m} \left[f_{w,b}(x^{(i)}) - y^{(i)} \right]^2 \quad \ldots \langle i \rangle$$

Gradient Descent Algorithm

① $w = w - \alpha \frac{\partial}{\partial w} J(w,b)$

② $b = b - \alpha \frac{\partial}{\partial b} J(w,b)$

① & ② are repeated until they converge.

Substituting <i> in ①'s derivative function

$$\frac{\partial}{\partial w} J(w,b) = \frac{\partial}{\partial w} \cdot \frac{1}{2m} \sum_{i=1}^{m} \left[f_{w,b}(x^{(i)}) - y^{(i)} \right]^2$$

$$\Rightarrow \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^{m} \left[wx^{(i)} + b - y^{(i)} \right]^2$$

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^{m} \frac{\partial}{\partial w} \left[wx^{(i)} + b - y^{(i)} \right]^2$$

{ Applying chain rule $\rightarrow \frac{d}{dx} P(q(x)) = P'(q(x)) \cdot \frac{d}{dx} q(x)$ }

$$\therefore \frac{1}{2m} \sum_{i=1}^{m} 2 \cdot [wx^{(i)} + b - y^{(i)}] \cdot \frac{\partial}{\partial w} (wx^{(i)} + b - y^{(i)})$$

$$\Rightarrow \frac{1}{m} \sum_{i=1}^{m} [wx^{(i)} + b - y^{(i)}] \cdot x^{(i)}$$

Writing in terms of $f_{w,b}(x)$,

$$\frac{1}{m} \sum_{i=1}^{m} [f_{w,b}(x^{(i)}) - y^{(i)}] x^{(i)} \quad \ldots\ldots (3)$$

<u>Substituting <i> in (2)'s derivative function</u>

$$\frac{\partial}{\partial b} J(w,b) = \frac{\partial}{\partial b} \cdot \frac{1}{2m} \sum_{i=1}^{m} [f_{w,b}(x^{(i)}) - y^{(i)}]$$

$$\Rightarrow \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^{m} [wx^{(i)} + b - y^{(i)}]^2$$

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^{m} \frac{\partial}{\partial b} [wx^{(i)} + b - y^{(i)}]^2$$

{Applying chain rule}

$$\therefore \frac{1}{2m} \sum_{i=1}^{m} 2 [wx^{(i)} + b - y^{(i)}] \cdot \frac{\partial}{\partial w} (wx^{(i)} + b - y^{(i)})$$

$$\Rightarrow \frac{1}{m} \sum_{i=1}^{m} [wx^{(i)} + b - y^{(i)}]$$

Writing in terms of $f_{w,b}(x)$,

$$\frac{1}{m} \sum_{i=1}^{m} [f_{w,b}(x^{(i)}) - y^{(i)}] \quad \ldots (4)$$

On substituting ③ ~~with~~ in ① and ④ in ② we get,

$$w = w - \alpha \cdot \frac{1}{m} \sum_{i=1}^{m} \left[f_{w,b}(x^{(i)}) - y^{(i)} \right] x^{(i)}$$

$$\& \; b = b - \alpha \cdot \frac{1}{m} \sum_{i=1}^{m} \left[f_{w,b}(x^{(i)}) - y^{(i)} \right]$$